Developing Elo Ratings for Club Football Matches

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Abstract

The Elo rating system is popular across many sports. It is relatively simple and transparent, with a logarithmic-curve formulation yielding both team strength ratings and predictions for future match-ups based on the results of previous matches. That simplicity though masks a great deal of scope for variation in the method. Additionally, the Elo-generated prediction does not yield a prediction for all potential outcomes in sports with a tied/drawn match outcome (like football). In this paper we consider some variations of the parameters in the Elo rating system, and evaluate them. We produce a tentative recommendation for the Elo rating system on this basis, and compare the resulting Elo ratings against other readily available and basic prediction methods.

JEL Classification: .
Keywords: Ratings, predictions, football.

1 Introduction

The Elo rating system (Elo, 1978) is popular in sporting contexts. The system generates a strength for each participant based on previous matches played. Each participant's strength is adjusted based on the difference between what happened in a match relative to would be expected to happen given the strengths of participants. It was developed for chess, an individual sport, and has been adopted in many sports, including team sports. One particular example is the FIFA rankings of international men's football teams.[1] It has many advantages; it is a simple method both in terms of computation, relying solely on match results, but also for understanding. A participant's rating is comparable to any other participant at a given point in time, enabling rankings to be formed. It is also internally coherent, generating forecasts that are then used to adjust the rating of participants in an 'error correction' style. It also allows for varying parameterisations as well as modifications of the basic method to allow for characteristics of a sport, such as the home advantage.

Such a system allows an answer to the common question posed by those interested in a sport: who is the best (Baker and McHale, 2015)? In principle, this question can be answered through time as well, since the format for evaluating performances does not change over time. A system that allows a

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[1]See https://inside.fifa.com/fifa-world-ranking/procedure-men.

rating for teams to be generated can be powerful. It can inform planning and preparation for the teams participating. It can influence sports league organisers when designing competitions. It can inform betting strategies, and indeed the pricing of bets by bookmakers.

An Elo rating system needs to be parameterised, with the key parameters determining how sensitive ratings are to each match result. With a high level of sensitivity, then in principle ratings adjust quickly to the *true strength* of a participant, but may then fluctuate substantially around that level. With a low level of sensitivity, there is less variation in ratings, but they may take longer to adjust to changes in a participant's true strength. In addition, a range of modifications to the methodology that alter how either ratings are updated or predictions are made might be employed. Kovalchik (2016) introduced a decay factor for the adjustment for each tennis player to allow early career player ratings to adjust more than later career. With Kovalchik (2016) the sensitivity is relative to career experience, but other systems such as FIFA's vary sensitivity by the type of match, with a World Cup final having a higher adjustment than a friendly match. Hvattum and Arntzen (2010) considered a range of parameters and also a modification to the method — they build in the goal difference from a match into the adjustment. Alongside the goal-difference augmentation, there are others that have been used in different implementations. A popular website, clubelo.com, adds a home advantage correction to allow the system to reflect that historically, home teams win more often than away teams for given relative strengths of the teams. Another modification is to allow a team's Elo rating to decay with the length of time between matches played.

Football presents an interesting environment for the Elo ratings for at a number of reasons. First, football already has a system of rating teams — position in league competitions and stage achieved in knock-out tournaments. Second, football teams usually compete within hierarchical league structures that only allow infrequent adjustment up and down the divisions. This can lead to anomalies where teams in lower divisions are better than teams in higher divisions. Thirdly, complementing the hierachical structure, football has knock-out cup tournaments that run alongside league competitions in which teams from different league divisions can face each other. Fourthly, such hierarchical league structures tend to operate on a pyramid basis where below a particular level, multiple regional divisions exist, within which quality differences may emerge in what are divisions intended to be of similar quality teams. Fifthly, rival hierarchical structures of leagues exist within close proximity of each other; throughout Europe each country operates its own league system. European cup competitions gives some sense of the relative quality differences across countries in Europe, but only amongst the elite clubs. An Elo rating system may allow a better comparison across countries. Sixthly, by and large, once created teams never die, unlike participants in, say, chess or tennis tournaments. This may create inflation in any Elo rating system, since a new team is allocated a notional Elo strength in points, points which are then distributed to other teams as competition ensues. This may mean that the same Elo rating value in, say, 1920 or 1970, may not mean the same thing in 2024.

Given the annual adjustment of teams via promotion and relegation, rating systems other than simple position within the league system, may be important for purposes such as recruitment. Because the mobility of labour is relatively high in football, clubs are continually needing to recruit players who are usually playing (or non-playing squad members) at another club, sometimes in a different division. A rating system can thus feed into recruitment decisions. This is particularly true where more than one league exists at the same level in a system, such as when competitions become more regional in their nature. For clubs able to recruit from other countries, the ability to rank teams using a system that in principle allows international comparisons between teams is important. One of the advantages of the

Elo rating system is that it only requires match results and the dates on which the matches occurred in order to be computed. In other words, league and cup matches, including those between teams in different divisions and countries, can equally be included in the rating system.

In this paper we evaluate a range of parameterisations and tweaks to the method. An ideal Elo rating parameterisation will allow comparisons both across clubs at a single point in time, and through time. We consider as much football as possible as a basis for evaluating the parameterisations and tweaks. Relative to Hvattum and Arntzen (2010) we also add in cup matches, which are often between teams of different divisions and hence act as an additional means to calibrate a system for more effective comparisons — in particular (but not only) across countries.

We focus our parameterisations on the adjustment factor: how much a team's Elo rating adjusts after a match outcome. While we could also vary the slope of the logarithmic curve through which Elo ratings are constructed, the same effect can be achieved by varying the adjustment factor over a larger range of values.

We consider a range of further tweaks to the Elo rating system. We vary the adjustment for each team based on two dimensions: first, how long the team has been in existence, and second how many matches have taken place in a season. First, we add a term for home advantage, reflecting that the home team wins more often than the away team, for two equal strength teams. Second, we increase the sensitivity of adjustment to cup matches. Third, we add a decay factor for teams that do not play for a length of time.

We evaluate these variations using a number of relatively common metrics: forecast error measures, forecast efficiency testing, and forecast encompassing testing alongside bookmaker prices (where available). We also consider whether trading strategies informed by Elo ratings would return a positive return on investment. In doing so we perform a relative evaluation of Elo predictions. We also address the issue of creating probabilities of distinct outcomes in football matches given the draw (tied) outcome exists.

We focus much of our attention on cup matches. We do evaluate using league matches, but the real added value for a rating system like Elo is being able to compare teams in different competitions. Ideally, the forecast performance would be equally effective in league and cup matches.

We find...

In Section **??** the relevant previous literature is reviewed, in Section 2 the modelling methodology adopted is set out, in Section 3 our dataset and sources are introduced, in Section 4 results from the econometric estimations are presented, and Section 5 concludes.

2 Methodology

We first outline the Elo rating system before outlining a number of adaptations, or tweaks, to the methodology. We then consider how ratings systems might be best evaluated.

2.1 The Elo rating system

Elo ratings were developed by (Elo, 1978) to use a logarithmic system to both rate participants, predict outcomes, and subsequently update ratings based on the difference between the prediction and the outcome. Elo ratings were originally used for chess. In the time since, they have been used across a wide range of sporting contexts due to their relative ease for computation. Hvattum and Arntzen (2010) considered their application to association football, where prediction methods have long been highly

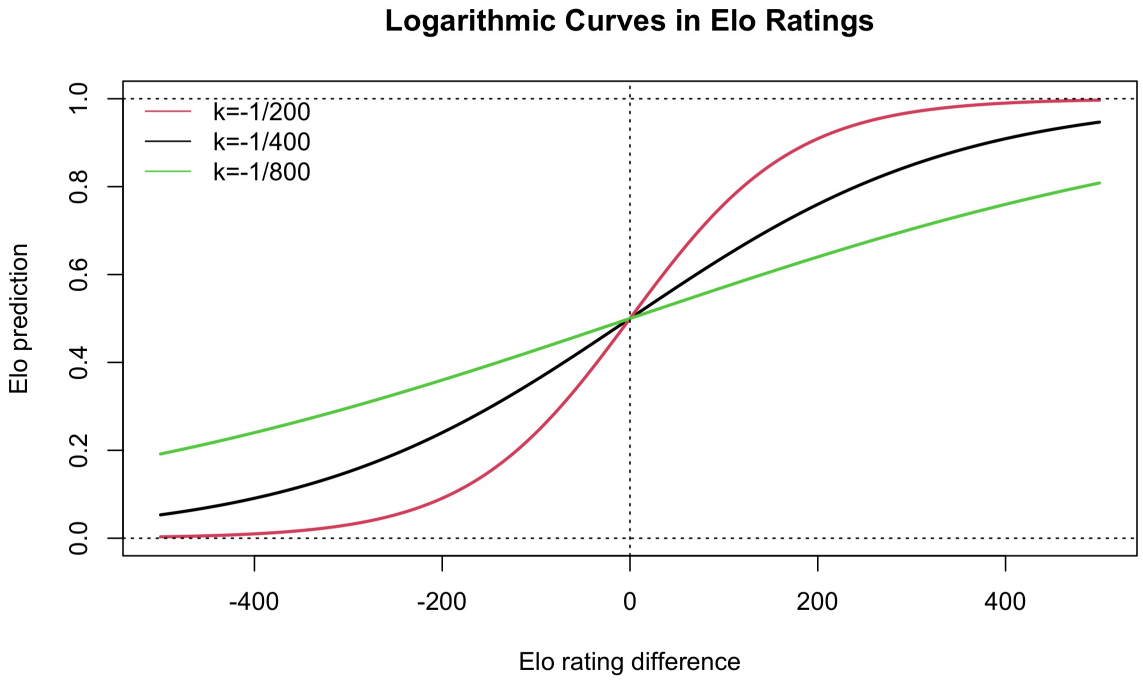


Figure 1: Conceptual plot of the logarithmic curve upon which Elo ratings are based.

valuable due to the popularity of gambling on football outcomes. They found that although Elo ratings were better than other statistical constructs for forecasting match outcomes they were not better than betting odds.

The Elo rating system assigns each participant an Elo rating. In an individual sport like chess, each person playing has their own Elo rating; in a team sport like football, a team will have its Elo rating. Each match has two teams, hence we can think about R_{it} and R_{jt} for the team i and j, at time t. The Elo prediction for such a match at time t, from team i's perspective, is constructed as:

$$E_{it} = \frac{1}{1 + d^{-k(R_{jt} - R_{it})}}. \tag{1}$$

Here, $d = 10$ is usually used, as is $k = -1/400$. The prediction from team j's perspective (E_{jt}) is defined similarly, and are such that $E_{it} + E_{jt} = 1$. Note that (1) is a logistic curve with base d, and hence is bound on the unit interval; that is, the prediction must lie between zero and one. The k parameter is one that can be adjusted, and the most basic description of the Elo system (via *Wikipedia*) has $k = -1/400$. In Figure 1 we plot what this means, conceptually, for the Elo rating system. Along the horizontal axis is the different in Elo ratings of two teams participating in a match, and on the vertical axis is the implied Elo prediction, calculated using (1). Hence the larger (smaller) is the denominator (hence smaller (larger) is k), the flatter (steeper) is the logarithmic curve, and hence the Elo prediction is less (more) extreme for any given difference in Elo ratings.

Then each team's Elo rating is adjusted once the match outcome S_{it} is known, where:

$$S_{it} = \begin{cases} 1 & \text{if team } i \text{ wins,} \\ 0.5 & \text{if game is a draw/tie,} \\ 0 & \text{if team } i \text{ loses.} \end{cases} \tag{2}$$

Again, an equivalent variable S_{jt} can be defined, and $S_{jt} = 1 - S_{it}$.

Then the adjustment to a team's rating is the difference between the predicted outcome (E_{it}) and the outcome (S_{it}), scaled by some factor K:

$$R_{it+1} = R_{it} + K(S_{it} - E_{it}). \tag{3}$$

Hence, Elo ratings will adjust more for a 'better' win; that is, if team i is expected to beat their opponents, such that $E_{it} > 0.5$, then $S_{it} - E_{it} < 0.5$ in the event team i wins. Conversely, if team i is not expected to win and hence $E_{it} < 0.5$, then $S_{it} - E_{it} > 0.5$ if they win.

The outcome variable we have defined in (2) has three possibilities: the home win, the away win, but also the draw (or tie). The prediction, E_{it}, does not factor in the possibility of a drawn outcome, but rather is a real number on the unit interval, ranging between certain defeat ($E_{it} = 0$) and certain victory ($E_{it} = 1$) in a contest. When the draw is a common outcome, as it is in football, it then becomes of interest to present the predictions from Elo ratings as probabilities of the three outcomes, and additionally commonplace to do so from the perspective of the home and away teams. Hence we denote $P(S_{it} = 1) = P(S_{jt} = 0)$ as the probability of the home win outcome, $P(S_{it} = 0.5) = P(S_{jt} = 0.5)$ for the draw and $P(S_{it} = 0) = P(S_{jt} = 1)$ for the away win, and seek estimators for these, which we denote $\widehat{P}_{elo}(S_{it} = 1)$ for the home win, to signify that it was generated from our Elo rating system.

One way to do this is to use a multinomial, or ordered, logit/probit model. Such a step naturally adds a complication in what is otherwise a very simple method to generate predictions, but provided that the estimated parameters of any such model are not changing by much through time or across competitions, then it could be applied more as a rule of thumb type adjustment to Elo predictions. Note also that (2) need not be the only way to quantify the outcomes of football matches; however, this definition is in-fitting with the Elo prediction scale from (1).

The adjustment factor K is the parameter most commonly varied between different applications of the Elo rating system. A smaller (larger) value of K will mean that Elo ratings adjust less (more) to match outcomes, conditional on the relative strengths of the two teams taking part.

By making a range of choices of K we thus evaluate different Elo rating schemes. We do not need to vary k however, because a range of combinations of k and K achieve the same set of ratings, and hence we can simply adjust K across a range of values.[2] We set as is convention, $k = -1/400$, and choose $K \in \{1, 5, 10, 20, 40\}$.

In this paper we attempt to determine the best choice of K, and to quantify the use of Elo ratings for football. The best choice of K will be the one that maximises forecast performance. While the Elo rating system could be compared to other rating systems, since there is no *true* rating system or set of rankings between teams, it is not clear what the benchmark would be to compare against. While league tables in league-type competitions might seem a natural comparison, they do have a number of

[2]We note that there is a relationship such that particular basic values of k and K yield identical ranking outcomes; that is, for example, $(k = -1/200, K = 10)$, $(k = -1/400, K = 20)$, and $(k = -1/800, K = 40)$. This is because both k and K relate to the sensitivity of the Elo system: K for ratings to outcomes relative to predictions, and k for predictions to Elo ratings.

drawbacks; they are based on relatively small samples of matches, and the constrain teams by divisions through seasons.

Alternatively, since Elo rating systems automatically produce predictions for matches based on information known before a match occurs, these predictions can be used to evaluate the system. If the forecasts are good, it implies that the rating for each team is accurate. Furthermore, other benchmark forecasts exist to rank Elo ratings against; a literature has emerged establishing the efficiency of bookmaker prices as predictors for matches (see, e.g. Croxson and Reade, 2014; Elaad et al., 2020). Finally, the existence of many hundreds of thousands of football matches across all types of competitions allows for a thorough evaluation that can cover particular subsets of matches of interest, for example cup matches, or matches at the start of a season.

2.2 Modifications to the Elo rating method

Alongside these parameterisations, we consider a number of modifications to the Elo methodology. We vary the adjustment factor for the experience of a team and on a seasonal basis, we add a home advantage parameter to reflect that the home team historically wins more often than the away team, we increase the weight on cup matches above that of league matches, and we add a rate of decay for teams not playing for a long time.

2.2.1 Exponentially declining adjustment factor

Varying the adjustment factor for the experience of teams is influenced by Kovalchik (2016) who specify a exponentially declining weight for the adjustment for each tennis player in response to a match outcome. The weight declines with the number of matches played by a player, and so we adapt this for teams to both account for the length of time a team has been in existence, but also the number of matches into a new season. The latter is designed to allow the rating system to adapt quickly to changes in the quality of teams between seasons due to recruitment.

Over a team's existence, we apply the following exponential function:

$$f(N) = 1 + 1.947453e^{-0.02662471N}, \tag{4}$$

where N is the number of games played in a team's history. This function is designed specifically such that it triples the baseline adjustment factor K in the team's first ever match, converges to 1 as $N \to \infty$ in order that the team's adjustment in the long term is K, and does almost all of that convergence before $N = 200$, as $f(N)$ is designed such that $f(200) = 1.01$.

Over a team's season, we apply the following exponential function:

$$f(N_s) = 1 + 1.389e^{-0.329N_s}, \tag{5}$$

where N_s is the number of matches a team has played in a season. This function is designed such that it doubles the baseline adjustment factor K in the team's first match of the season, converges to 1 as $N \to \infty$, and is essentially 1 by the 15th match of the season.

It is worth emphasising that this mofication ensures that the Elo rating system is not zero sum, since the adjustments for the two teams involved can be different, rather than equal but of the opposite sign. Furthermore, doubling (tripling) adjustment factors for teams early in their season (existence) will add inflation into the system in terms of the individual ratings of teams.

2.2.2 Home advantage

The home advantage method frequently employed adds some Elo points, denoted H, into the prediction calculation in (1), such that it becomes:

$$E_{it} = \frac{1}{1 + d^{-k(R_{jt} - R_{it} - H)}}. \tag{6}$$

We choose H to reflect recent trends in the home advantage. If, for example, home wins occur 45% of the time and away wins 30% of the time, this implies that $E(S_{it}) = 0.45 \times 1 + 0.25 \times 0.5 + 0.3 \times 0 = 0.575$ from (2). We then seek that (6) is equal to 0.575 in expectation, to understand what the home advantage adjustment should be. We assume that $R_{jt} - R_{it} = 0$ to consider the case of two equally balanced teams, and solve:

$$E_{it} = \frac{1}{1 + 10^{-H/400}} = 0.575, \tag{7}$$

which yields $H = 52.6$. We take for each club in our dataset the moving average of S_{it} from their most recent 30 home matches and use that to calculate what their home advantage parameter should be, and add that in to (6).

2.2.3 Inactivity Penalty

Our dataset includes teams that exist only for a short time period before folding (e.g. Wanderers, Peterborough and Fletton United), and others who disappear before reforming and rising again (e.g. Accrington, Aldershot, Wimbledon). In order to account for this, we employ a decay factor for a team's Elo rating. This factor is designed to fall very slowly at first; after 100 (365) days without a match, a team will still have 99.9% (98.8%) of its Elo rating after its most recent match. Between one and three calendar years, however, the team will lose almost half (45%) of its most recent Elo rating, and will remain at this level (a baseline of about 800). When calculating Elo predictions even within seasons, we apply this decay factor for each team participating. The rationale is that a team that may suffer from gaps between performances.

We employ the following formula:

$$R(t) = R_{infty} + \frac{R_0 - R_\infty}{1 + e^{m(t - t_0)}}, \tag{8}$$

where R_0 is the rating after the club's last match, R_∞ is the ratings floor (800), t is the number of days since the last match, t_0 is the inflection point (which we take to be 730 days or two years), and m is a steepness parameter (we set to $m = 0.01$). This formula is arbitrarily designed; ideally it would be based on actual patterns of decay in performance for delays between recorded matches.

2.3 Alternative Prediction Methods

There exist a vast array of potential methods for forecasting football matches, and Elo ratings are but one. A commonly studied source of forecasts are bookmaker prices. On a more basic level, league table information can be employed for making forecasts. For a match within a league, a basic rule of thumb might be that a team in a higher league position would be more likely to win. A more sophisticated version of this might take into account the point in the season (by using points per game perhaps),

and may also take into account shorter term form. A cup match between teams in different divisions might make use of the full position in the league ladder to emphasise divisional differences. We employ a league-table basic forecast as a baseline for comparing our Elo forecasts against.

We can generate a league-based forecast using a Bradley-Terry type formulation, which like the Elo prediction generates a number between zero and one. That is, if we take league position for team i (j) as pos_{it} (pos_{jt}) then the prediction would be $p_{ijt}^{pos} = pos_{it}/(pos_{it} + pos_{jt})$, and similarly for form. Hence for a team in 1st place playing a team in 24th place, the forecast would be that the team in 1st place wins 96% of the time, and the team bottom 4%.

2.4 Evaluation of the ratings

We use a number of methods to determine this 'optimal' choice of parameters. We consider:

1. Basic forecast error measures.
2. Calibration testing, or Mincer-Zarnowitz regression testing.
3. Hypothetical betting returns.
4. Eye-ball tests: we look at plots of series and other descriptive information such as ranks of teams alongside league positions. This is a 'sniff test', or a plausibility test of the rankings.

Where betting data exists for matches, we make use of that as a comparison with the range of Elo ratings we produce. In addition, we use two commonly used sources of Elo ratings: those produced by clubelo.com and the Opta Power Rankings.[3] The *clubelo* system adds modifications based on goals scored, on the goal difference in a match, and has a time-varying home advantage factor. It uses a standard adjustment factor of $K = 20$, and has a system to deal with two-legged cup matches, but does not otherwise weight cup matches differently to league matches. The Opta Power Rank is based on an Elo system that adds more weight to cup and international matches by having separate Elo ratings for leagues, countries and continents, and which adds in the goal difference in a match result to the adjustment.[4]

Our external comparison is important as a relative metric; *clubelo* produces a set of ratings for the public according to a method that is outlined on its website. Bookmakers produce forecasts for matches with the purpose of profit maximising and private enterprises. Both are effective at their purpose, and if the ratings produced here are comparable a high bar threshold has been reached. Equally, failing to reach that level is not necessarily a reason to reject ratings produced. The purpose of this research exercise is to evaluate a range of modifications to the basic Elo method, some of which are employed by clubelo, and some of the ideas will be employed by bookmakers in contructing their odds. The utility or otherwise of such modifications is helpful for the research community.

[3]The systems are details here (clubelo): clubelo.com/System, and here (Opta Power Rankings): theanalyst.com/articles/power-rankings-your-club-ranked. Both were downloaded from github.com/tonyelhabr/club-rankings?tab=readme-ov-file.

[4]That is, an FA Cup match between a Premier League team and a Championship team will lead to an adjustment for the two teams involved, as well as the league Elo ratings, and each club's total Elo rating is the sum of its own (within-league) rating, its division Elo rating, its country and its continent Elo rating. Games in the Club World Cup would affect the club Elo rating and the continental Elo rating.

2.4.1 Forecast Errors

The forecast error is defined as the difference between a forecast and the outcome. Using the notation introduced in this paper we can write:

$$FE_{ijt} = S_{ijt} - E_{ijt}. \tag{9}$$

Forecast errors naturally require an event to have occurred, and its outcome used to calculate the error. The simple mean of (9) over a number of events would be the mean forecast error (MFE):

$$MFE = \frac{1}{K}\sum_{k=1}^{K} FE_{ijt}. \tag{10}$$

The mean squared forecast error, or Brier score, ensures that positive and negative errors do not cancel each other out, and provides a measure of the variance of forecast errors:

$$MSFE = \frac{1}{K}\sum_{k=1}^{K} FE_{ijt}^2. \tag{11}$$

2.4.2 Regression-based testing

The second evaluation method we use is to run a Mincer and Zarnowitz (1969) regression, which allows us to test for the efficiency of a forecast. An efficient forecast is one that incorporates all available information at the time the forecast was made. This implies that the forecast error (9) is unpredictable; that is, it is mean zero with a constant variance. If this is the case, then we could write that $S_{ijt} - E_{ijt} = u_{ijt}$, where $u_{ijt} \sim (0, \sigma_u^2)$, where σ_u^2 is the variance of u_{ijt} and 0 is the mean. Then if the following regression model is run:

$$S_{ijt} = \alpha + \beta E_{ijt} + u_{ijt}, \qquad u_{ijt} \sim (0, \sigma_u^2), \tag{12}$$

an efficient forecast is such that $\alpha = 0$ and $\beta = 1$. In using this method to evaluate forecasts, we would choose values for adjustment paramater K and decide on other modifications based on which produced forecasts that generated coefficient estimates for (12) that were closest to $\alpha = 0$ and $\beta = 1$.

The evaluation method in (12) implies a more graphical method to judge forecasts. Forecasts can be aggregated into intervals on the unit interval, and evaluated based on how often such forecasts occur. For example, forecasts at 10% (rounded to the nearest integer) could be evaluated based on how often the event in question occurred; if the event occurs 10% of the time, and the same was true for all other forecast values, the forecast would be said to be well calibrated. This can be graphically represented by plotting such forecasts against the frequency with which they occur. A well calibrated, or efficient forecast, would be one where all points are collected around the 45-degree line in such a plot.

More broadly, equation (12) can be used as a basis to evaluate competitor forecasts. A rearrangement yields:

$$FE_{ijt} = \gamma + \delta E_{ijt} + v_{ijt}, \qquad v_{ijt} \sim (0, \sigma_v^2), \tag{13}$$

where the principle is that no information regarding the outcome should be of use in explaining the errors from the forecast; that is, $\delta = 0$ for efficiency, as well as $\gamma = 0$. This approach can be broadened

to consider information from other sources. If we consider some vector of potential predictor variables for the match outcome, X_{ijt}, we could add them into the model to yield:

$$FE_{ijt} = \gamma + \delta E_{ijt} + \theta X_{ijt} + e_{ijt}, \qquad e_{ijt} \sim (0, \sigma_e^2), \tag{14}$$

and the hypothesis of forecast efficiency remains one of all the parameters being equal to zero, hence being insignificant for explaining the errors in the Elo predictions.

Built upon this approach is the encompassing approach to considering different forecasts for the same event (Fair and Shiller, 1989; Chong and Hendry, 1986). If we have an alternative forecast, say from bookmakers, then if that is B_{ijt}, and the associated forecast error is FE^B_{ijt}, then we would insert the forecast into (14) rather than X_{ijt}:

$$FE_{ijt} = \gamma_e + \theta_e B_{ijt} + e_{ijt}, \qquad e_{ijt} \sim (0, \sigma_e^2), \tag{15}$$

Here, if $\theta \neq 0$ then the bookmaker forecast is able to explain some of the forecast error from our Elo ratings; that is, it can explain something about the outcome that our Elo forecast couldn't. It is important to apply this method both ways, and hence run:

$$FE^B_{ijt} = \gamma_b + \theta_b E_{ijt} + \varepsilon_{ijt}, \qquad \varepsilon_{ijt} \sim (0, \sigma_\varepsilon^2). \tag{16}$$

Here, if $\theta_b \neq 0$ then Elo ratings help explain some of the outcomes that bookmaker prices are unable to explain. However, if $\theta_e = 0$ or $\theta_b = 0$ then one of these forecasts is unable to explain the other. In these cases, forecasts are said to *encompass* each other. In particular:

- $\theta_e = 0$ and $\theta_b \neq 0$: Bookmaker forecasts cannot explain Elo forecast errors, and Elo forecasts can explain bookmaker forecast errors. Elo forecasts *encompass* bookmaker forecasts.
- $\theta_e \neq 0$ and $\theta_b = 0$: Elo forecasts cannot explain bookmaker forecast errors, and bookmaker forecasts can explain Elo forecast errors. Bookmaker forecasts *encompass* Elo forecasts.
- $\theta_e = 0$ and $\theta_b = 0$: Bookmaker forecasts cannot explain Elo forecast errors, and Elo forecasts cannot explain bookmaker forecast errors.
- $\theta_e \neq 0$ and $\theta_b \neq 0$: Bookmaker forecasts can explain Elo forecast errors, and Elo forecasts can explain bookmaker forecast errors. Neither encompass each other. A linear combination of forecasts may perform better than either individually.

2.4.3 Hypothetical Betting Returns

As our alternative candidate forecasts are bookmaker prices as well as two publicly constructed sets of Elo ratings, we use these in our encompassing approach, but also, we consider whether any of these sets of Elo ratings can produce a profitable betting strategy.

The expected return from a bet can be written, where a bettor stakes a dollar, D are the decimal odds, and P^* is the true probability of the event occurring, as:

$$E(R) = P^* D - 1. \tag{17}$$

Hence, if $P^* = 1/D$, then the expected return is zero. This gives an expression commonly used to interpret decimal odds of an event: the reciprocal of them is the *implied probability* of the event occurring.

If a bookmaker offers odds such that $P^* > 1/D$ then $E(R) > 0$ and positive expected returns can be made.

Constructing any such test requires a number of assumptions. Here, we assume for simplicity that if $\widehat{P}_{elo} > D$ that a bet is placed.

A betting rule must be established, to set out the circumstances under which a bet would be placed. The simplest betting strategy assumes a bet is placed if $P > 1/D$. Slightly more sophisticated strategies might employ the Kelly criterion.

2.4.4 Types of matches to be used

The value of using football matches is the huge number of matches that occur every day. This enables us to both look at all of these matches to determine whether, as a whole, Elo ratings appear to forecast well. It also enables us to consider in some more detail certain subsets of matches.

For example, we are interested in matches involving teams from different divisions — some cup matches. With a hierarchical system like the English Football League, the fact that teams are in divisions for a season before being promoted or relegated between divisions means that for some time, there may be better teams at the top of a lower division than the teams at the bottom of the division above.

The test thus of whether a choice of K leads to too much variation in team Elo ratings such that between-division comparisons are misleading, or whether it allows a sufficient amount of variation such that between-division comparisons are accurate. If for matches between teams of different divisions, Elo ratings provide efficient forecasts, it suggests that the choice of K is optimal.

In addition, matches at the beginning of a season tend to be hard to predict. Teams will have recruited between seasons, and some teams will have been promoted or relegated. Our exponential decay modification is designed to mitigate this problem by increasing the weight at the start of the season, allowing team Elo ratings to adjust more quickly at that time. This also means that a metric for whether that modification has helped would be to focus analysis just on early season matches.

2.4.5 Different forecasts

As mentioned earlier, Elo ratings are designed for contests where no tied outcome is possible. In football, tied outcomes are commonplace, and as such this presents a problem in evaluating forecasts. The Elo forecast itself can be thought of as the probability that team i wins if a draw is not possible; the other event, that team j wins, is the remainder of the probability space as it is the only other event. If a draw is possible, however, this adds a third event in the event space and neither probability generated by Elo ratings takes this into account.

While forecast errors can still be calculated according to (9), using the outcome varable defined in (2), then strictly speaking this is not an appropriate comparison since the Elo prediction does not allow for the draw, which the outcome variable treats at the value 0.5. An Elo prediction of 0.5 would imply a draw is likely, but does not give a particuar probability.

One alternative is to evaluate the Elo prediction E_{it} against a home-win outcome variable:

$$H_{it} = \begin{cases} 1 & \text{if (home) team } i \text{ wins,} \\ 0 & \text{otherwise.} \end{cases} \tag{18}$$

Again, though, this is not strictly speaking the right comparison, since E_{it} is the probability of the home win when ignoring the possibility of a drawn outcome.

We thus create probabilities for the three outcomes in a football match to evaluate Elo forecasts. A model like an ordered logistic regression can transform each Elo prediction into probabilities, but it would need to be estimated over the matches taking place *before* those it is being used to predict. That is, a training and a test sample would be needed. Otherwise, future information would be used to transform the Elo prediction into probabilities.

A rule of thumb could be used to convert the Elo prediction into probabilities, or a regression method like an ordered logistic regression could be employed. We adopt the latter strategy, and use a rolling sample period for the purposes. We estimate an ordered logit on a rolling five-year training period, and project forecasts forward. As such, we are always forecasting one year ahead as our procedure rolls forward by a year at a time. The logit model is simply the outcome variable for a match on the Elo prediction for that match.

Our ordered logit regression can be written as:

$$P(y = y) \tag{19}$$

3 Data

Our data covers league and cup football since the late 1800s, mostly sourced from Soccerbase. While it seems unlikely that using matches in the 1880s can improve Elo ratings in the 2020s, it allows a historical appraisal of the strengths of teams throughout the history of football, as well as not imposing any arbitrary dates upon where to start using matches from.

This is also relevant as over more recent years, Soccerbase has included more and more results from around the world. Because Soccerbase has only added those countries in recent years, the records are thus quite incomplete, we are particularly interested in the impact of increasing a team's adjustment factor for its first few years of existence, as per our exponentially declining adjustment explained in Section 2.2.1. It seems possible that in adding this feature, variable start dates for different teams will matter less.

An important advantage of using Elo ratings is the ability to compare teams across different, non-hierarchical competitions. That is, to be able to rank, say, teams in England with teams in France or Germany. Opta Power Rankings achieve this by calculating separate Elo ratings for clubs within divisions, divisions, countries and continents, which are then summed up to give a team's Elo rating. We employ as one of our proposed variants a conceptually similar idea: we increase the weight placed on all cup matches — domestic and international. Again, the existence of domestic and international cup matches is a particular test of the effectiveness of the Elo rating method as well as our proposed modifications.

The first match in our database took place in 1866 and was a friendly match between Nottingham Forest and Notts County. In 1871 the first FA Cup match took place, a knock-out competition that runs to this day, and is regarded to be instrumental in the development of football. There were undoubtedly other regional cup competitions that started prior to the FA Cup, but their records are not on Soccerbase, and hence we only include them to the extent that they appear in the results listed in a number of different football club archives.[5]

[5]We use results published in Oldham Athletic's history books, and on websites run by fans of Macclesfield, Luton Town,

It is important to note that any Elo rating system is as good as the data that is inputted into it; while those data requirements are minimal (teams, result, date), they can still dictate how effective Elo ratings can be at particular points in time and in particular places.[6] The first Football League match took place in September 1888, and again other attempts at league competitions started at around this point, many of which either failed or were merged into the Football League or 'non-league' system (see, e.g. Che and Humphreys, 2015).

We collect clubelo ratings from github.com/xgabora/Club-Football-Match-Data/, which also provides bookmaker prices from www.football-data.co.uk/.

Table 1 presents summary statistics of the dataset. Across the 913,266 matches, the team listed first (usually the home team) scores 1.7 goals, and the team listed second (usually the away team) scores 1.2. Assuming the first team listed is always the home team, the home team wins 48.3% of the time, and the away team 27.8% of the time. The remainder of matches, 23.9%, were drawn.

Table 1: Summary Statistics [old, to be updated]

Statistic	N	Mean	St. Dev.	Min	Max
goals1	497,046	1.761	1.491	0	36
goals2	497,046	1.227	1.199	0	22
outcomeH	497,046	0.488	0.500	0	1
outcomeD	497,046	0.237	0.425	0	1
outcomeA	497,046	0.275	0.447	0	1
league	497,046	0.837	0.369	0	1
england	497,046	0.753	0.431	0	1
After 2002-08-01					
goals1	160,997	1.564	1.348	0	16
goals2	160,997	1.278	1.211	0	13
outcome==H	160,997	0.441	0.497	0	1
outcome==D	160,997	0.237	0.425	0	1
outcome==A	160,997	0.322	0.467	0	1
Bet365 P(H)	67,549	0.433	0.133	0.041	0.914
Bet365 P(D)	67,558	0.263	0.031	0.000	0.407
Bet365 P(A)	67,558	0.304	0.122	0.000	0.866
league	160,997	0.764	0.425	0	1
england	160,997	0.748	0.434	0	1

4 Results

4.1 Establishing choice of adjustment factor

We use forecast error measures and Mincer-Zarnowitz regression tests to evaluate different adjustment factors alongside clubelo ratings and bookmaker prices. In the first instance we consider bias via mean

Exeter City amongst others.

[6]Because of this incompleteness of the data available to us, we have posted our datasets on Github in order that interested readers may suggest to add matches that are not already in our database, and to correct any errors that are in our dataset as it stands.

forecast errors in Table 2. Here, we restrict attention to just the matches where we have clubelo predictions and betting prices to also evaluate, in order to ensure a 'fair' comparison over the same set of matches. In practice this restricts us to matches since 2000 in the leagues covered by www.football-data.co.uk. In Table 18 in the Appendix on page 30 presents the table with all matches, since equally the value of calculating ratings ourselves is being able to use *all* matches to evaluate rather than just the ones that clubelo provides predictions for.

Our table is split into three panels, separated by horizontal lines. In the top panel we compare the Elo prediction against the outcome variable in equation (2), namely the variable that includes the draw outcome. In the middle panel, we consider the Elo predictions against the home-win outcome, hence treating the draw and away win as the same outcome. In the bottom panel we present the Elo predictions transformed into probabilities of the home win, draw and away win.

When calculating using the outcome variable including draws, varying the K adjustment factor makes a small difference, increasing the mean forecast error from 0.079 for $K = 1$ to 0.083 for $K = 40$. These small positive biases suggest that, on average, basic Elo ratings predict the draw and home win to happen more frequently than they actually happen.

We can alternatively consider the mean forecast error where we only evaluate against the home win outcome; in other words, our outcome variable is 1 only when the home team wins. There is no draw equivalent here, and the away equivalent would be the reciprocal, and hence we only list for the home win outcome. Here, our choice of K decreases the mean forecast error from -0.053 ($K = 1$) down to -0.049 ($K = 40$). This is a somewhat mechanical difference, since the difference in the calculation is that now the draw outcome is represented by a 0 whereas above it was represented by 0.5. Hence there are a proportion observations where the error is more negative by 0.5 due to only considering the home win outcome.

Our Elo-based forecasts transformed into probabilities for each outcome are in the bottom panel, with '.PH.' signifying the probability of the home outcome. These errors are smaller than the plain Elo rating predictions, with the base at $K = 10$ yielding -0.004, 0.026 and -0.022 for the home, draw and away win events respectively. The differences between different adjustment factors are at the second and third decimal place here, which is negligible.

In Table 3 we consider mean squared forecast errors from equation (11). The table is organised similarly to Table **??**. As opposed to bias, MSFEs more readily evaluate variation in forecasts. A better forecast has a smaller bias and lower variation. Considering basic Elo predictions and the ternary outcome variable (2), $K = 10$ is marginally better than other adjustment factors, with an MSFE of 0.165, but the difference is negligible. When evaluating just against the home win outcome, $K = 10$ and $K = 20$ achieve the lowest MSFE, and the difference between these Ks and others is larger here than it was for (2).

When evaluating Elo ratings transformed into home, draw and away win probabilities, all are small, and increasing the K factor reduces each MSFE, but only at the third decimal place.

In Table 4 we present Mincer-Zarnowitz regression tests for our different adjustment factors using the ternary outcome variable from (2). We plot the implied slopes in Figure 2. The constant term is significant for all adjustment factors and hence all lie above the 45-degree line in the plot. This reflects that the outcome variable includes the draw outcome though. Nonetheless, the closest slope to one is delivered by an adjustment factor $K = 10$.

In Table 5 we present Mincer-Zarnowitz regression tests for our different adjustment factors using the home-win outcome variable. We plot the implied slopes in Figure 3. The constant term is now much

Table 2: Mean forecast error measures for Elo predictions, varying the K adjustment factor.

Statistic	N	Mean	St. Dev.	Min	Max
elo.base1.error	68,428	0.079	0.405	−0.802	0.796
elo.base5.error	68,428	0.080	0.400	−0.890	0.892
elo.base10.error	68,428	0.080	0.399	−0.916	0.912
elo.base20.error	68,428	0.081	0.400	−0.928	0.936
elo.base40.error	68,428	0.083	0.404	−0.952	0.954
elo.base1.H.error	68,428	−0.053	0.482	−0.802	0.796
elo.base5.H.error	68,428	−0.052	0.477	−0.898	0.892
elo.base10.H.error	68,428	−0.052	0.475	−0.921	0.912
elo.base20.H.error	68,428	−0.051	0.476	−0.936	0.936
elo.base40.H.error	68,428	−0.049	0.479	−0.953	0.954
elo.base1.PH.error	68,428	−0.006	0.481	−0.921	0.944
elo.base1.PD.error	68,428	0.033	0.439	−0.265	0.949
elo.base1.PA.error	68,428	−0.027	0.438	−0.866	0.969
elo.base5.PH.error	68,428	−0.004	0.476	−0.883	0.921
elo.base5.PD.error	68,428	0.028	0.439	−0.272	0.926
elo.base5.PA.error	68,428	−0.024	0.435	−0.802	0.952
elo.base10.PH.error	68,428	−0.004	0.475	−0.867	0.910
elo.base10.PD.error	68,428	0.026	0.439	−0.274	0.917
elo.base10.PA.error	68,428	−0.022	0.434	−0.774	0.947
elo.base20.PH.error	68,428	−0.003	0.476	−0.846	0.894
elo.base20.PD.error	68,428	0.023	0.439	−0.276	0.904
elo.base20.PA.error	68,428	−0.020	0.434	−0.743	0.940
elo.base40.PH.error	68,428	−0.002	0.478	−0.814	0.878
elo.base40.PD.error	68,428	0.022	0.439	−0.276	0.886
elo.base40.PA.error	68,428	−0.020	0.436	−0.703	0.930

Table 3: Mean squared forecast errors

Statistic	N	Mean	St. Dev.	Min	Max
elo.base1.error2	68,428	0.171	0.118	0.000	0.644
elo.base5.error2	68,428	0.166	0.132	0.000	0.795
elo.base10.error2	68,428	0.165	0.138	0.000	0.840
elo.base20.error2	68,428	0.166	0.147	0.000	0.876
elo.base40.error2	68,428	0.170	0.159	0.000	0.910
elo.base1.H.error2	68,428	0.235	0.075	0.037	0.644
elo.base5.H.error2	68,428	0.230	0.111	0.008	0.806
elo.base10.H.error2	68,428	0.229	0.124	0.004	0.848
elo.base20.H.error2	68,428	0.229	0.138	0.001	0.877
elo.base40.H.error2	68,428	0.232	0.156	0.0004	0.910
elo.base1.PH.error2	68,428	0.231	0.142	0.002	0.892
elo.base1.PD.error2	68,428	0.194	0.236	0.002	0.901
elo.base1.PA.error2	68,428	0.193	0.189	0.001	0.939
elo.base5.PH.error2	68,428	0.227	0.145	0.005	0.849
elo.base5.PD.error2	68,428	0.194	0.231	0.005	0.858
elo.base5.PA.error2	68,428	0.189	0.187	0.002	0.906
elo.base10.PH.error2	68,428	0.226	0.144	0.007	0.828
elo.base10.PD.error2	68,428	0.193	0.229	0.007	0.840
elo.base10.PA.error2	68,428	0.189	0.186	0.002	0.898
elo.base20.PH.error2	68,428	0.226	0.144	0.009	0.798
elo.base20.PD.error2	68,428	0.193	0.227	0.008	0.817
elo.base20.PA.error2	68,428	0.189	0.186	0.003	0.884
elo.base40.PH.error2	68,428	0.228	0.142	0.013	0.771
elo.base40.PD.error2	68,428	0.193	0.226	0.012	0.785
elo.base40.PA.error2	68,428	0.191	0.186	0.004	0.865

smaller, and indeed insignificantly different from zero for $K = 20$. For $K = 1$ the line deviates most from the 45-degree line, and $K = 10$ and $K = 20$ are closest to it.

Table 4: Regression output of Mincer-Zarnowitz calibration test regressions on different adjustment factors, using ternary outcome variable.

	Dependent variable:				
	outcome				
K	1	5	10	20	40
	(1)	(2)	(3)	(4)	(5)
Constant	−0.232***	0.026***	0.090***	0.151***	0.220***
	(0.004)	(0.002)	(0.002)	(0.002)	(0.001)
Elo prediction	1.656***	1.140***	1.013***	0.892***	0.757***
	(0.008)	(0.004)	(0.004)	(0.003)	(0.003)
Observations	798,607	798,607	798,607	798,607	798,607
R^2	0.048	0.075	0.085	0.092	0.091
Adjusted R^2	0.048	0.075	0.085	0.092	0.091
Residual Std. Error (df = 798605)	0.416	0.410	0.407	0.406	0.406
F Statistic (df = 1; 798605)	40,016.470***	64,363.010***	74,583.150***	80,776.280***	79,634.930***
Note:				*p<0.1; **p<0.05; ***p<0.01	

In Table 6 we present Mincer-Zarnowitz regression results with our Elo predictions converted into a probability of a home win using our ordered logit model in (19), against the home win outcome, and Figure 3 plots this graphically. The constant coefficients are all negligible and mainly insignificant, while the slope coefficients are close to one. The plots are all very close to the 45-degree line also. This suggests that the process of converting Elo predictions into probabilities using a regression approach is effective at removing biases and improving forecasts, on average.

Table 5: Regression output of Mincer-Zarnowitz calibration test regressions on different adjustment factors, using home-win outcome variable and the basic Elo prediction.

	Dependent variable:				
	outcomeH				
	(1)	(2)	(3)	(4)	(5)
K	1	5	10	20	40
Constant	−0.409***	−0.134***	−0.066***	0.0001	0.073***
	(0.005)	(0.003)	(0.002)	(0.002)	(0.002)
Elo prediction	1.773***	1.224***	1.088***	0.957***	0.812***
	(0.010)	(0.005)	(0.004)	(0.004)	(0.003)
Observations	798,607	798,607	798,607	798,607	798,607
R^2	0.040	0.062	0.072	0.077	0.076
Adjusted R^2	0.040	0.062	0.072	0.077	0.076
Residual Std. Error (df = 798605)	0.489	0.484	0.481	0.480	0.480
F Statistic (df = 1; 798605)	33,071.720***	53,187.190***	61,540.590***	66,544.000***	65,572.260***
Note:					*p<0.1; **p<0.05; ***p<0.01

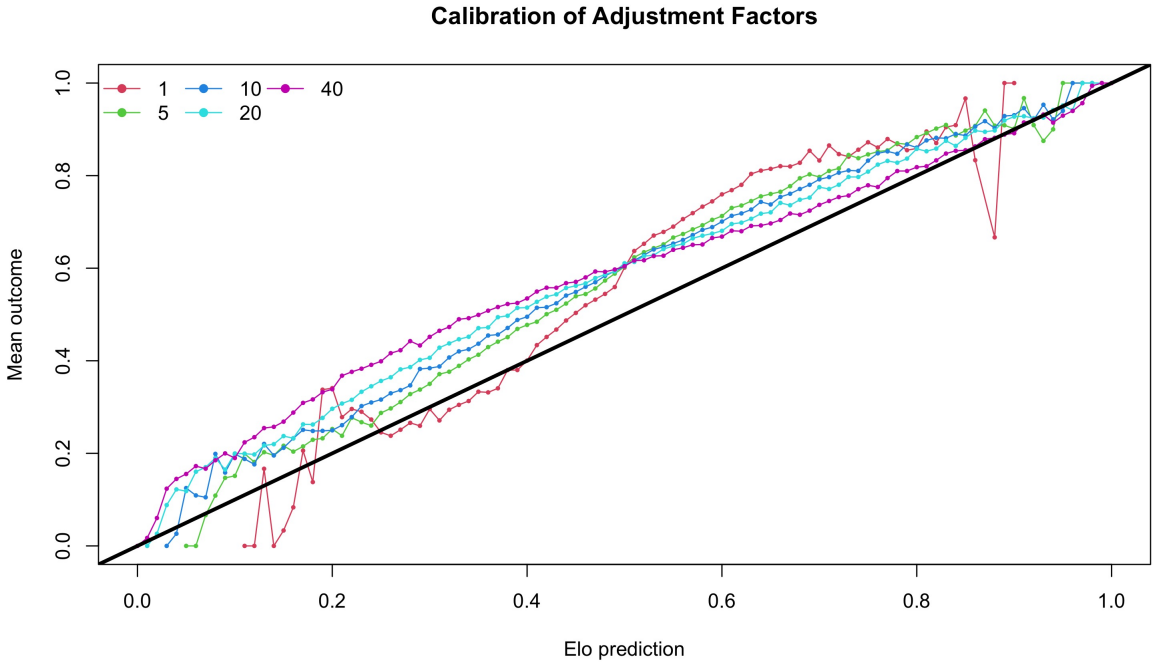


Figure 2: Calibration of adjustment factors with ternary outcome variable.

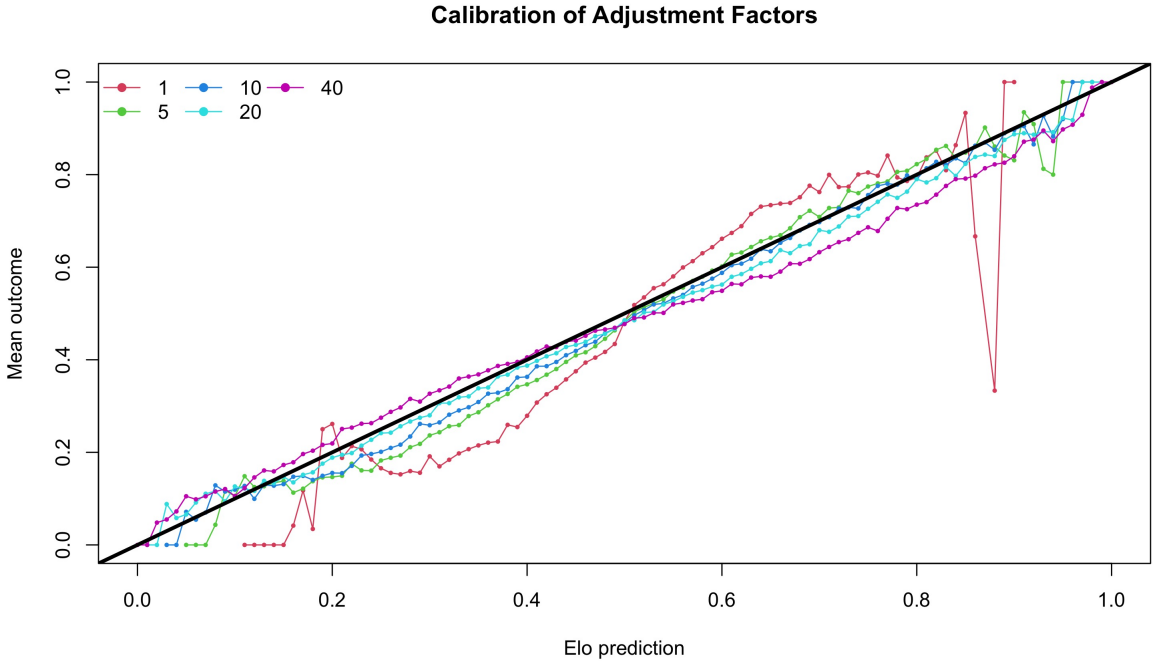


Figure 3: Calibration of adjustment factors with Elo prediction converted into a home-win probability, and home-win outcome variable.

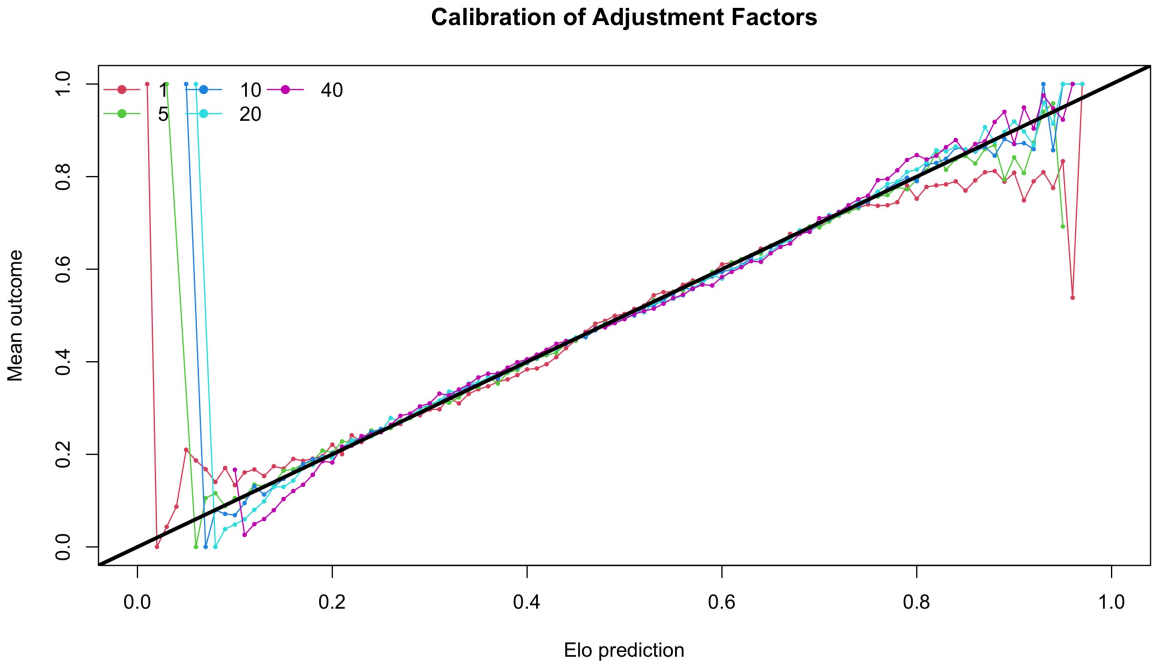


Figure 4: Calibration of adjustment factors with Elo prediction converted into a home-win probability, and home-win outcome variable.

Table 6: Regression output of Mincer-Zarnowitz calibration test regressions on different adjustment factors, using home-win outcome variable and Elo generated outcome probabilities.

	Dependent variable:				
	outcomeH				
	(1)	(2)	(3)	(4)	(5)
K	1	5	10	20	40
Constant	−0.001	0.002	0.004*	0.006***	0.007***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Elo prediction	0.998***	0.991***	0.988***	0.984***	0.981***
	(0.005)	(0.004)	(0.004)	(0.004)	(0.004)
Observations	798,281	798,281	798,281	798,281	798,281
R^2	0.049	0.070	0.079	0.084	0.083
Adjusted R^2	0.049	0.070	0.079	0.084	0.083
Residual Std. Error (df = 798279)	0.487	0.482	0.479	0.478	0.478
F Statistic (df = 1; 798279)	41,114.350***	60,516.650***	68,603.260***	73,239.970***	71,867.050***
Note:					*p<0.1; **p<0.05; ***p<0.01

Summarising the range of evaluations included in this section for the choice of K, we conclude that a choice of $K = 10$ is optimal. While true team strengths will change, and hence a higher K factor may be helpful for detecting these, equally there are lots of matches over which for team strengths to adjust, and as such, a smaller K factor appears slightly favourable.

4.2 Evaluating Modifications

In Table 7 we consider mean forecast errors from our various modified forecasts, taking $K = 10$ as the adjustment factor. The table is organised as Table 2.

Employing the exponential decay method for club existence and seasons reduces the MFE a tiny bit further relative to the $K = 10$ baseline, and this is lower than the equivalent errors for the clubelo method.

There is almost no difference across all modifications except the home advantage adjustment, which reduces the mean error to 0.018. An interpretation of this would be that the error in all other Elo predictions is primarily driven by the home advantage, where home teams win more often than their relative strength suggests they should.

Indeed, it is a comparable mean error level to bookmaker forecasts for the home win, albeit not for the draw or away win (0.006, -0.0005, -0.005). This is true also for clubelo forecasts, which are as effective for draw and away forecasts as home ones.

In this case, our various adjustments appear to do quite little for the mean forecast error, but the home advantage modification increases it.

In Table 8 we present mean squared forecast errors. The table is formatted and ordered as for MFEs. By and large, the same conclusions are upheld for standard Elo predictions, with our Elo baseline, clubelo and our modifications all delivering similar mean squared errors. The home advantage correction helps for the basic comparison of the ternary outcome variable but hinders for the home win outcome, as before. When converting the Elo predictions into probabilities for the three outcomes, almost all differences between methods vanish, with the home MSFE being about 0.23, and the draw and away outcomes at about 0.19. This is slightly larger than bookmakers.

The comparison of basic Elo predictions with the home-win outcome yields larger MSFEs, and this is true also for the home-win probabilities also. When comparing to the general outcome variable, the error will automatically be more variable since if a draw occurs and the prediction is 0.75, the general outcome variable is 0.5, but the home-win outcome variable is 0. To give a sense of scale in percentage point terms, a squared forecast error of 0.2 equates to 0.45 hence 45 percentage points. This is large, but with binary outcome variables, this is also to be expected. In addition, the difference between the MSFE for $K = 10$ of 0.22 (47 percentage points) and the bookmaker at 0.26 (51 percentage points) is relatively small.

4.3 Evaluating Selected Elo Ratings Against Other Predictors

Our main focus to now has been selecting the best parameterisation for Elo ratings; of the adjustment factors, $K = 10$ is our preferred choice, while of the modifications, the exponential decay and cup adjustments have encouraging properties in the types of matches where baseline Elo ratings struggle [DO THIS!].

In this section, we compare our favoured Elo forecast method against a range of league table generated forecasts, clubelo forecasts, and bookmaker prices. We employ the same methods as in the last two

Table 7: Mean forecast error measures for Elo predictions, employing a range of modifications to the method.

Statistic	N	Mean	St. Dev.	Min	Max
elo.kovalchik.dbl.error	68,428	0.081	0.399	−0.920	0.919
elo.decay.error	68,428	0.080	0.399	−0.886	0.890
elo.home.adv.error	68,428	0.018	0.406	−0.920	0.958
elo.goal.diff.error	68,428	0.082	0.404	−0.978	0.979
elo.cup.error	68,428	0.080	0.400	−0.933	0.936
elo.full.error	68,428	0.023	0.410	−0.978	0.992
elo.kovalchik.dbl.H.error	68,428	−0.051	0.475	−0.927	0.919
elo.decay.H.error	68,428	−0.052	0.476	−0.899	0.890
elo.home.adv.H.error	68,428	−0.114	0.483	−0.920	0.958
elo.goal.diff.H.error	68,428	−0.050	0.479	−0.978	0.979
elo.cup.H.error	68,428	−0.052	0.476	−0.943	0.936
elo.full.H.error	68,428	−0.109	0.485	−0.980	0.992
elo.kovalchik.dbl.PH.error	68,428	−0.003	0.475	−0.858	0.903
elo.kovalchik.dbl.PD.error	68,428	0.023	0.439	−0.276	0.911
elo.kovalchik.dbl.PA.error	68,428	−0.020	0.434	−0.760	0.941
elo.decay.PH.error	68,428	−0.003	0.477	−0.819	0.873
elo.decay.PD.error	68,428	0.023	0.439	−0.268	0.891
elo.decay.PA.error	68,428	−0.020	0.435	−0.707	0.922
elo.home.adv.PH.error	68,428	−0.006	0.482	−0.747	0.892
elo.home.adv.PD.error	68,428	0.026	0.440	−0.267	0.850
elo.home.adv.PA.error	68,428	−0.019	0.438	−0.745	0.898
elo.goal.diff.PH.error	68,428	−0.003	0.476	−0.805	0.868
elo.goal.diff.PD.error	68,428	0.023	0.439	−0.277	0.880
elo.goal.diff.PA.error	68,428	−0.020	0.434	−0.688	0.927
elo.cup.PH.error	68,428	−0.004	0.476	−0.873	0.913
elo.cup.PD.error	68,428	0.025	0.439	−0.275	0.920
elo.cup.PA.error	68,428	−0.022	0.435	−0.781	0.949
elo.full.PH.error	68,428	−0.006	0.479	−0.727	0.858
elo.full.PD.error	68,428	0.022	0.440	−0.273	0.839
elo.full.PA.error	68,428	−0.016	0.436	−0.675	0.888

Table 8: Mean squared forecast errors

Statistic	N	Mean	St. Dev.	Min	Max
elo.kovalchik.dbl.error2	68,428	0.166	0.140	0.000	0.847
elo.decay.error2	68,428	0.166	0.136	0.000	0.791
elo.home.adv.error2	68,428	0.165	0.152	0.000	0.918
elo.goal.diff.error2	68,428	0.170	0.171	0.000	0.958
elo.cup.error2	68,428	0.166	0.141	0.000	0.877
elo.full.error2	68,428	0.168	0.179	0.000	0.985
elo.kovalchik.dbl.H.error2	68,428	0.229	0.127	0.002	0.859
elo.decay.H.error2	68,428	0.229	0.118	0.005	0.809
elo.home.adv.H.error2	68,428	0.246	0.156	0.001	0.918
elo.goal.diff.H.error2	68,428	0.231	0.177	0.0001	0.958
elo.cup.H.error2	68,428	0.229	0.128	0.002	0.888
elo.full.H.error2	68,428	0.247	0.200	0.00004	0.985
elo.kovalchik.dbl.PH.error2	68,428	0.226	0.140	0.008	0.816
elo.kovalchik.dbl.PD.error2	68,428	0.193	0.227	0.008	0.830
elo.kovalchik.dbl.PA.error2	68,428	0.189	0.186	0.002	0.885
elo.decay.PH.error2	68,428	0.227	0.119	0.014	0.762
elo.decay.PD.error2	68,428	0.194	0.227	0.012	0.794
elo.decay.PA.error2	68,428	0.190	0.178	0.005	0.850
elo.home.adv.PH.error2	68,428	0.232	0.122	0.010	0.796
elo.home.adv.PD.error2	68,428	0.194	0.230	0.019	0.723
elo.home.adv.PA.error2	68,428	0.192	0.188	0.009	0.806
elo.goal.diff.PH.error2	68,428	0.226	0.146	0.015	0.754
elo.goal.diff.PD.error2	68,428	0.193	0.227	0.014	0.775
elo.goal.diff.PA.error2	68,428	0.189	0.186	0.005	0.858
elo.cup.PH.error2	68,428	0.227	0.146	0.006	0.833
elo.cup.PD.error2	68,428	0.193	0.229	0.006	0.847
elo.cup.PA.error2	68,428	0.189	0.187	0.002	0.901
elo.full.PH.error2	68,428	0.230	0.129	0.019	0.737
elo.full.PD.error2	68,428	0.194	0.227	0.025	0.704
elo.full.PA.error2	68,428	0.190	0.189	0.011	0.789

Table 9: Regression output of Mincer-Zarnowitz calibration test regressions on different modifications, using home-win outcome variable.

	Dependent variable:					
	outcome					
	(1)	(2)	(3)	(4)	(5)	(6)
	Expo	Decay	Home	GD	Cup	All
Constant	0.126***	0.180***	0.145***	0.254***	0.105***	0.255***
	(0.002)	(0.002)	(0.002)	(0.001)	(0.002)	(0.001)
Elo prediction	0.943***	0.835***	0.769***	0.687***	0.985***	0.592***
	(0.003)	(0.004)	(0.003)	(0.002)	(0.004)	(0.002)
Observations	799,963	799,963	799,963	799,963	799,963	799,963
R^2	0.091	0.056	0.065	0.097	0.088	0.086
Adjusted R^2	0.091	0.056	0.065	0.097	0.088	0.086
Residual Std. Error (df = 799961)	0.406	0.414	0.412	0.405	0.407	0.407
F Statistic (df = 1; 799961)	80,476.230***	47,518.480***	55,269.400***	85,820.310***	77,427.870***	75,525.930***
Note:						*p<0.1; **p<0.05; ***p<0.01

Table 10: Regression output of Mincer-Zarnowitz calibration test regressions on different modifications, using home-win outcome variable.

	Dependent variable:					
	outcomeH					
	(1)	(2)	(3)	(4)	(5)	(6)
	Expo	Decay	Home	GD	Cup	All
Constant	−0.028***	0.031***	−0.001	0.109***	−0.050***	0.116***
	(0.002)	(0.002)	(0.002)	(0.001)	(0.002)	(0.002)
Elo prediction	1.012***	0.896***	0.816***	0.739***	1.057***	0.627***
	(0.004)	(0.005)	(0.004)	(0.003)	(0.004)	(0.003)
Observations	799,963	799,963	799,963	799,963	799,963	799,963
R^2	0.076	0.047	0.053	0.081	0.074	0.070
Adjusted R^2	0.076	0.047	0.053	0.081	0.074	0.070
Residual Std. Error (df = 799961)	0.480	0.488	0.486	0.479	0.481	0.482
F Statistic (df = 1; 799961)	66,254.300***	39,403.390***	44,604.880***	70,956.120***	63,838.250***	60,590.940***
Note:						*p<0.1; **p<0.05; ***p<0.01

Table 11: Regression output of Mincer-Zarnowitz calibration test regressions on different modifications, using home-win outcome variable.

	Dependent variable:					
	outcomeH					
	(1)	(2)	(3)	(4)	(5)	(6)
	Expo	Decay	Home	GD	Cup	All
Elo prediction	0.985***					
	(0.004)					
elo.decay.PH		0.992***				
		(0.005)				
elo.home.adv.PH			0.994***			
			(0.005)			
elo.goal.diff.PH				0.982***		
				(0.004)		
elo.cup.PH					0.987***	
					(0.004)	
elo.full.PH						0.989***
						(0.004)
Constant	0.005***	0.002	0.003	0.007***	0.004**	0.004**
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Observations	799,635	799,635	799,635	799,635	799,635	799,635
R^2	0.083	0.055	0.054	0.088	0.081	0.072
Adjusted R^2	0.083	0.055	0.054	0.088	0.081	0.072
Residual Std. Error (df = 799633)	0.478	0.486	0.486	0.477	0.479	0.481
F Statistic (df = 1; 799633)	72,815.930***	46,274.740***	45,590.140***	77,531.920***	70,734.670***	62,441.540***
Note:						*p<0.1; **p<0.05; ***p<0.01

sections, and hence begin with bias via mean forecast errors, variance via mean squared forecast errors, efficiency via Mincer-Zarnowitz testing, and informativeness using encompassing.

In Table 12 we present our measure of bias, the mean squared forecast error.

The clubelo ratings have a mean error of 0.079, and our preferred Elo adjustments had a mean error of about 0.8. League table information delivers slightly smaller mean error terms.

Considering the home-win outcome, all measures are relatively similar in yielding a small bias of around 5–6 percentage points.

Converting league-based forecasts into probabilities yields small biases — again this appears to be an effective way to generate unbiased forecasts.

In Table 13 we present our measure of variation in forecasts, the mean squared forecast error. The clubelo forecasts produce identical MSFEs as our own Elo forecasts, and these are quite a bit smaller than any of the league-table based forecasts; while they are unbiased, they are more volatile around the eventual outcome.

Table 12: Mean forecast error measures for alternatives to Elo predictions.

Statistic	N	Mean	St. Dev.	Min	Max
pos.share.error	21,103	0.070	0.525	−0.960	0.960
form.share.error	20,756	0.085	0.436	−1.148	1.275
full.position.share.error	21,103	0.075	0.502	−0.952	0.952
elo.clubelo.error	68,428	0.079	0.399	−0.923	0.934
pos.share.H.error	21,103	−0.060	0.589	−0.960	0.960
form.share.H.error	20,756	−0.046	0.507	−1.148	1.275
full.position.share.H.error	21,103	−0.055	0.568	−0.952	0.952
elo.clubelo.H.error	68,428	−0.053	0.475	−0.933	0.934
elo.clubelo.PH.error	68,428	−0.005	0.475	−0.831	0.891
elo.clubelo.PD.error	68,428	0.001	0.439	−0.305	0.887
elo.clubelo.PA.error	68,428	0.004	0.434	−0.709	0.942
B365PH.error	68,426	0.006	0.469	−0.908	0.949
B365PD.error	68,428	−0.0005	0.438	−0.607	0.937
B365PA.error	68,428	−0.005	0.428	−0.851	0.972
pos.share.PH.error	21,103	−0.045	0.485	−0.657	0.672
pos.share.PD.error	21,103	0.014	0.439	−0.267	0.806
pos.share.PA.error	21,103	0.031	0.447	−0.406	0.851
form.share.PH.error	20,756	−0.046	0.492	−0.697	0.742
form.share.PD.error	20,756	0.011	0.439	−0.269	0.821
form.share.PA.error	20,756	0.035	0.451	−0.488	0.874

In Table 14 we report the Mincer-Zarnowitxz regression tests on our alternative forecast models. The clubelo predictions are closest to the 45-degree line but are still some way off; Tables **??** and 9 display the equivalent regressions for our Elo forecasts and show that by this evaluation, our Elo ratings are better calibrated than clubelo, and much more so than the league-table based forecasts.

Table 13: Mean squared forecast errors

Statistic	N	Mean	St. Dev.	Min	Max
elo.clubelo.error2	68,428	0.166	0.150	0.000	0.872
elo.clubelo.H.error2	68,428	0.229	0.145	0.001	0.872
elo.clubelo.PH.error2	68,428	0.226	0.143	0.010	0.794
elo.clubelo.PD.error2	68,428	0.193	0.208	0.009	0.788
elo.clubelo.PA.error2	68,428	0.188	0.201	0.002	0.887
B365PH.error2	68,426	0.220	0.140	0.001	0.900
B365PD.error2	68,428	0.192	0.205	0.000	0.877
B365PA.error2	68,428	0.184	0.194	0.000	0.945
pos.share.error2	21,103	0.281	0.251	0.0001	0.922
form.share.error2	20,756	0.197	0.194	0.000	1.626
full.position.share.error2	21,103	0.257	0.209	0.00000	0.907
pos.share.H.error2	21,103	0.350	0.238	0.002	0.922
form.share.H.error2	20,756	0.259	0.194	0.000	1.626
full.position.share.H.error2	21,103	0.325	0.171	0.002	0.907
pos.share.PH.error2	21,103	0.237	0.078	0.108	0.451
pos.share.PD.error2	21,103	0.193	0.222	0.038	0.649
pos.share.PA.error2	21,103	0.200	0.211	0.022	0.725
form.share.PH.error2	20,756	0.244	0.063	0.092	0.550
form.share.PD.error2	20,756	0.193	0.220	0.031	0.674
form.share.PA.error2	20,756	0.205	0.217	0.017	0.764
lg.PH.error2	20,756	0.222	0.134	0.012	0.787
lg.PD.error2	20,756	0.192	0.226	0.005	0.844
lg.PA.error2	20,756	0.187	0.210	0.001	0.908

Hence, even if league based forecasts aren't hugely volatile, and aren't biased, they aren't well calibrated. A league-based forecast of 20% won't occur 20% of the time.

Table 14: Regression output of Mincer-Zarnowitz calibration test regressions on alternative forecast models, using ternary outcome variable.

	Dependent variable:			
	outcome			
	(1)	(2)	(3)	(4)
	clubelo	Position	Form	Full Position
Constant	0.161***	0.453***	0.473***	0.190***
	(0.004)	(0.002)	(0.002)	(0.003)
Elo prediction	0.842***	0.324***	0.290***	0.846***
	(0.008)	(0.003)	(0.004)	(0.006)
Observations	88,754	282,913	249,169	280,033
R^2	0.115	0.030	0.016	0.060
Adjusted R^2	0.115	0.030	0.016	0.060
Residual Std. Error	0.397 (df = 88752)	0.413 (df = 282911)	0.414 (df = 249167)	0.406 (df = 280031)
F Statistic	11,554.190*** (df = 1; 88752)	8,816.788*** (df = 1; 282911)	4,174.996*** (df = 1; 249167)	17,800.040*** (df = 1; 280031)
Note:				*p<0.1; **p<0.05; ***p<0.01

Table 15: Regression output of Mincer-Zarnowitz calibration test regressions on alternative forecast models, using home-win outcome variable.

	Dependent variable:			
	outcomeH			
	(1)	(2)	(3)	(4)
	clubelo	Position	Form	Full Position
Constant	0.001	0.314***	0.337***	0.023***
	(0.005)	(0.002)	(0.003)	(0.004)
Elo prediction	0.900***	0.357***	0.313***	0.932***
	(0.009)	(0.004)	(0.005)	(0.008)
Observations	88,754	282,913	249,169	280,033
R^2	0.095	0.026	0.013	0.051
Adjusted R^2	0.095	0.026	0.013	0.051
Residual Std. Error	0.473 (df = 88752)	0.493 (df = 282911)	0.497 (df = 249167)	0.487 (df = 280031)
F Statistic	9,282.806*** (df = 1; 88752)	7,476.666*** (df = 1; 282911)	3,373.341*** (df = 1; 249167)	15,024.850*** (df = 1; 280031)
Note:				*p<0.1; **p<0.05; ***p<0.01

We finally present an encompassing analysis, in order to determine which, if any, forecasts may encompass others, and hence which forecasts may be redundant. In Table 17 we present a matrix that contains information regarding the encompassing regressions involving all of our forecasts.

With this adjustment, the magnitude of forecast errors in terms of variability is similar to that of bookmaker odds.

As we have introduced betting prices as an alternative forecast here, it makes sense to further consider whether a betting strategy using any of our candidate forecasts could deliver a profitable return. In doing so, however, we do not find that any of our candidate forecasts can deliver anything even close to a positive return.

Table 16: Regression output of Mincer-Zarnowitz calibration test regressions on alternative forecast models, using home-win outcome variable and probability of home win forecasts.

	Dependent variable:				
	outcomeH				
	(1)	(2)	(3)	(4)	(5)
elo.clubelo.PH	0.970***				
	(0.010)				
pos.PH		0.982***			
		(0.011)			
form.PH			0.967***		
			(0.017)		
full.position.PH				1.041***	
				(0.008)	
lg.PH					1.027***
					(0.008)
Constant	0.009*	0.009	0.017**	−0.019***	−0.013***
	(0.005)	(0.006)	(0.008)	(0.004)	(0.004)
Observations	87,518	282,913	249,169	280,033	249,169
R^2	0.095	0.026	0.013	0.054	0.056
Adjusted R^2	0.095	0.026	0.013	0.054	0.056
Residual Std. Error	0.473 (df = 87516)	0.493 (df = 282911)	0.497 (df = 249167)	0.486 (df = 280031)	0.486 (df = 249167)
F Statistic	9,137.010*** (df = 1; 87516)	7,472.170*** (df = 1; 282911)	3,406.205*** (df = 1; 249167)	16,047.840*** (df = 1; 280031)	14,723.370*** (df = 1; 249167)
Note:					*p<0.1; **p<0.05; ***p<0.01

Table 17: Regression output of Mincer-Zarnowitz calibration test regressions on alternative forecast models, using home-win outcome variable and probability of home win forecasts.

	Dependent variable:				
	outcomeH				
	(1)	(2)	(3)	(4)	(5)
Observations	87,518	282,913	249,169	280,033	249,169
R^2	0.095	0.026	0.013	0.054	0.056
Adjusted R^2	0.095	0.026	0.013	0.054	0.056
Residual Std. Error	0.473 (df = 87516)	0.493 (df = 282911)	0.497 (df = 249167)	0.486 (df = 280031)	0.486 (df = 249167)
F Statistic	9,137.010*** (df = 1; 87516)	7,472.170*** (df = 1; 282911)	3,406.205*** (df = 1; 249167)	16,047.840*** (df = 1; 280031)	14,723.370*** (df = 1; 249167)
Note:					*p<0.1; **p<0.05; ***p<0.01

4.4 Evaluating over different types of matches

A final evaluation to be carried out is to consider particular subsets of matches that might prove problematic for forecasting. Matches early in a season, after which teams have recruited and may have changed their true strength, are one type. Cup matches where a team faces an opponent from a different league (or even country) are another.

5 Conclusions

In this note we explore Elo ratings for their applicability in the context of hierarchical league systems such as those that exist in football. We find that the optimal choice of K is 10, with K larger for the start of a club's existence, each season, and for cup matches.

The biggest improvement in forecasting though comes from employing an ordered logit model to convert measures on the unit interval into probabilities for the home win, draw and away win.

A Extra Results

B Illustrating the Method: A Team Example

In Table 19 we present the first ten fixtures of Oldham Athletic as recorded on www.soccerbase.com, from the year the club was admitted to the Football League, 1907. In the preceding season, the club had reached the First Round of the FA Cup for the first time, and hence appeared in recognised results sources for the first time.[7]

As the match on January 12 1907 was the first recognised match, Oldham had an Elo rating of 1000, the initial value given to new entrant teams. Kidderminster, at this point, had played one previous match — a heavy defeat, 13-0, to Darwen, in the FA Cup in 1891, and hence their Elo rating was 990.5 for the Oldham match. This yielded an Elo prediction of 0.51. Of course, in all likelihood Kidderminster played many matches between 1891 and 1907, just as Oldham had done, but these aren't recorded. The Elo rating system is only as good as the information supplied.

Oldham won that match 5-0, and hence their Elo rating increased to 1014.6 before their next round match, on February 2 1907, against Liverpool — an already well established Football League team with an Elo rating of 1211.8. Football League members since 1893, this was their 490th recorded match. This disparity yielded an Elo prediction of 0.24. In probability terms, this was about a 24% chance of an Oldham win, a 24% chance of a draw, and a 51% chance of a Liverpool win. Liverpool won 1-0, and hence Oldham's Elo rating adjusted back down — but not back to 1000, since the defeat was expected and so the adjustment smaller.

Infrequent matches will may mean that Elo ratings are less effective. The one-goal margin of the result against Liverpool suggests that Oldham were perhaps a little stronger than their Elo rating suggested, but without the results Oldham were achieving in Lancashire competition up to February 1907, their Elo rating may not be that close to the team's actual strength. Equally, with the additional matches

[7]The club was formed in 1895 and had played in leagues in Manchester and Lancashire up to the point of joining the Football League in 1907. In principle all results for all local league competitions could be incorporated into a grand Elo rating scheme, but this would be a non-trivial undertaking. A remarkable and comprehensive source of results like www.thefootballarchives.com/network/home.php would be especially well suited to such an endeavour; its owner, however, refuses to allow researchers to use the data.

Table 18: Mean forecast error measures for different Elo forecasts (plus bookmakers).

Statistic	N	Mean	St. Dev.	Min	Max
elo.kovalchik.dbl.PH.error	796,299	−0.002	0.478	−0.942	0.917
elo.kovalchik.dbl.PD.error	796,299	0.001	0.424	−0.414	0.974
elo.kovalchik.dbl.PA.error	796,299	0.001	0.434	−0.792	0.970
elo.decay.PH.error	796,299	−0.002	0.486	−0.891	0.884
elo.decay.PD.error	796,299	0.001	0.424	−0.404	0.931
elo.decay.PA.error	796,299	0.001	0.440	−0.725	0.962
elo.home.adv.PH.error	796,299	−0.001	0.486	−0.826	0.959
elo.home.adv.PD.error	796,299	0.001	0.425	−0.412	0.944
elo.home.adv.PA.error	796,299	−0.001	0.440	−0.923	0.920
elo.goal.diff.PH.error	796,299	−0.002	0.477	−0.928	0.944
elo.goal.diff.PD.error	796,299	0.001	0.424	−0.412	0.967
elo.goal.diff.PA.error	796,299	0.001	0.433	−0.744	0.972
elo.cup.PH.error	796,299	−0.002	0.479	−0.948	0.919
elo.cup.PD.error	796,299	0.001	0.424	−0.416	0.964
elo.cup.PA.error	796,299	0.001	0.434	−0.820	0.976
elo.full.PH.error	796,299	−0.001	0.481	−0.827	0.878
elo.full.PD.error	796,299	0.001	0.424	−0.424	0.926
elo.full.PA.error	796,299	−0.0001	0.435	−0.681	0.920
elo.base1.PH.error	796,299	−0.002	0.487	−0.962	0.988
elo.base1.PD.error	796,299	0.001	0.425	−0.419	0.976
elo.base1.PA.error	796,299	0.001	0.441	−0.930	0.989
elo.base5.PH.error	796,299	−0.002	0.482	−0.953	0.968
elo.base5.PD.error	796,299	0.001	0.424	−0.414	0.975
elo.base5.PA.error	796,299	0.001	0.437	−0.849	0.987
elo.base10.PH.error	796,299	−0.002	0.479	−0.944	0.947
elo.base10.PD.error	796,299	0.001	0.424	−0.413	0.972
elo.base10.PA.error	796,299	0.001	0.435	−0.846	0.984
elo.base20.PH.error	796,299	−0.002	0.478	−0.940	0.935
elo.base20.PD.error	796,299	0.001	0.424	−0.413	0.971
elo.base20.PA.error	796,299	0.001	0.434	−0.835	0.979
elo.base40.PH.error	796,299	−0.002	0.478	−0.947	0.902
elo.base40.PD.error	796,299	0.001	0.424	−0.413	0.976
elo.base40.PA.error	796,299	0.001	0.434	−0.795	0.971
elo.clubelo.PH.error	87,518	−0.005	0.473	−0.837	0.892
elo.clubelo.PD.error	87,518	−0.0001	0.437	−0.305	0.892
elo.clubelo.PA.error	87,518	0.005	0.432	−0.713	0.942
elo.kovalchik.dbl.error	796,625	0.097	0.406	−0.952	0.971
elo.kovalchik.dbl.H.error	796,625	−0.022	0.480	−0.962	0.971
elo.kovalchik.dbl.D.error	796,625	−0.262	0.453	−0.989	0.963
elo.kovalchik.dbl.A.error	796,625	−0.215	0.505	−0.989	0.991
elo.decay.error	796,625	0.097	0.414	−0.916	0.927
elo.decay.H.error	796,625	−0.022	0.488	−0.924	0.927
elo.decay.D.error	796,625	−0.262	0.446	−0.964	0.922
elo.decay.A.error	796,625	−0.214	0.491	−0.964	0.958
elo.home.adv.error	796,625	0.099	0.413	−0.973	0.968

Kidderminster would have played prior to their cup match with Oldham, again the Elo prediction might have been different.

	Home	Goals		Away	Elo ratings			Probabilities		
Date	team	Home	Away	team	Home	Prediction	Away	Home	Draw	Away
1907-01-12	Oldham	5	0	Kidderminster	1000.0	0.51	990.5	50.2	24.6	25.3
1907-02-02	Oldham	0	1	Liverpool	1014.6	0.24	1211.8	24.3	24.3	51.4
1907-09-07	Stoke	1	3	Oldham	1097.7	0.63	1007.3	61.9	20.8	17.3
1907-09-14	Oldham	2	1	West Brom	1019.8	0.35	1124.7	33.9	26.2	39.9
1907-09-21	Bradford	1	0	Oldham	1036.1	0.50	1032.8	49.2	24.8	26.0
1907-09-28	Oldham	3	0	Hull	1022.9	0.49	1028.1	47.9	25.1	27.0
1907-10-05	Derby	1	0	Oldham	1110.8	0.61	1033.0	60.2	21.4	18.4
1907-10-12	Oldham	4	0	Lincoln	1029.1	0.64	926.3	63.5	20.1	16.3
1907-10-19	Fulham	1	2	Oldham	1096.2	0.59	1032.7	58.2	22.2	19.7
1907-10-26	Oldham	1	0	Barnsley	1038.6	0.52	1024.5	50.9	24.4	24.7

Table 19: Elo ratings for Oldham Athletic's first ten recognised football matches in 1907.

In September 1907 Oldham were admitted to the Football League, and thus began playing regularly in competitions for which there exists comprehensive, widely available information.[8] They played four matches in September, and another four in October. Each match is listed in Table 19 and Oldham's Elo rating as it adjusted, can be tracked. By their final listed match, a 1-0 win over Barnsley, Oldham's Elo rating was 1038.6.

The course of Oldham's Elo rating throughout the remainder of that first season is plotted in Figure 5. A blue line connects the club's Elo rating before each match, and each match is denoted by a symbol. A blue square means Oldham won that match, a green circle they drew, and a red triangle means they lost. Hence after their first win, their Elo rating increased, before that Liverpool defeat brought a shift down in their Elo rating. Once in the FL, Oldham won more matches (24) than they lost (12), and hence by the end of the time period, their Elo rating is higher still than in Table 19, at 1069.3. Many of the adjustments once Oldham were in the league are smaller than others; this reflects that Oldham faced teams with Elo ratings both better (e.g. Fulham) and worse (e.g. Lincoln). Marked on Figure 5 with dark blue diamond symbols is the league position of the club before each match; a lower number is a higher position in the league table, and hence the Elo rating and league position are negatively correlated.

In Figure 6 the Elo ratings for the entire history of Oldham Athletic since (and including) 1907 are plotted, again with league position in darkblue.[9] The club's initial ascent up to the abandonment of football due to the First World War is clear, with an Elo rating of over 1200 achieved, and the club finished that final season before the war in second place in the FL. A subsequent decline took the club to its lowest ebb in the 1950s and 1960s, before two quick promotions in the early 1970s saw the club occupy the Second Division for almost twenty years. The late 1980s and early 1990s saw the club's second most successful era on the field, with three cup runs of note (FA Cup Semi Finals twice, League Cup final) and promotion to the First Division (which became the Premier League in 1992). At this time, a record high Elo rating of above 1300 was achieved.

A sharp fall in the late 1990s saw the club fall back two divisions into the third tier (now known as League One), where the club was until 2018, when it fell into the fouth tier, and in 2022 out of the

[8]Though of course, friendly matches would still not be included in the historical records and hence would be matches not used for the calculation of Elo ratings.

[9]Note that this league position is calculated out of all teams in league competitions. For example, if Oldham were first in the second division, and the first division had 20 teams, this would yield a league position of 21.

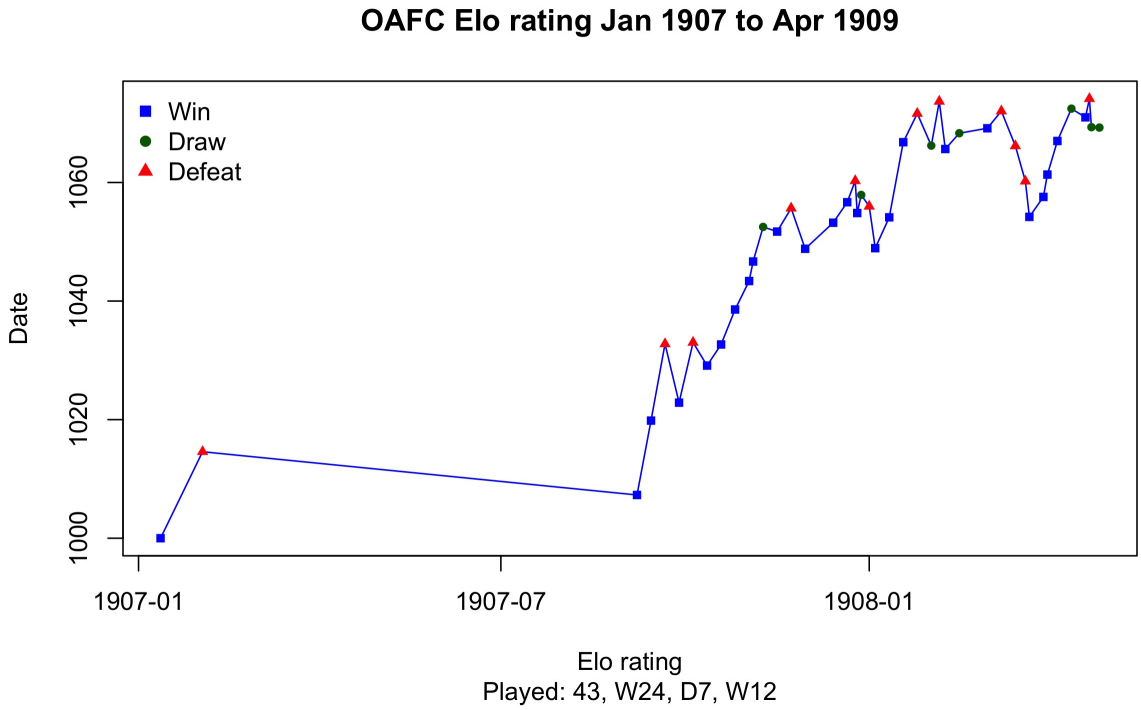


Figure 5: Oldham Athletic's Elo rating, January 1907 through to April/May 1909. A line connects the Elo ratings between different matches. A blue square indicates that match was a win for Oldham, a green circle a draw, and a red triangle a defeat.

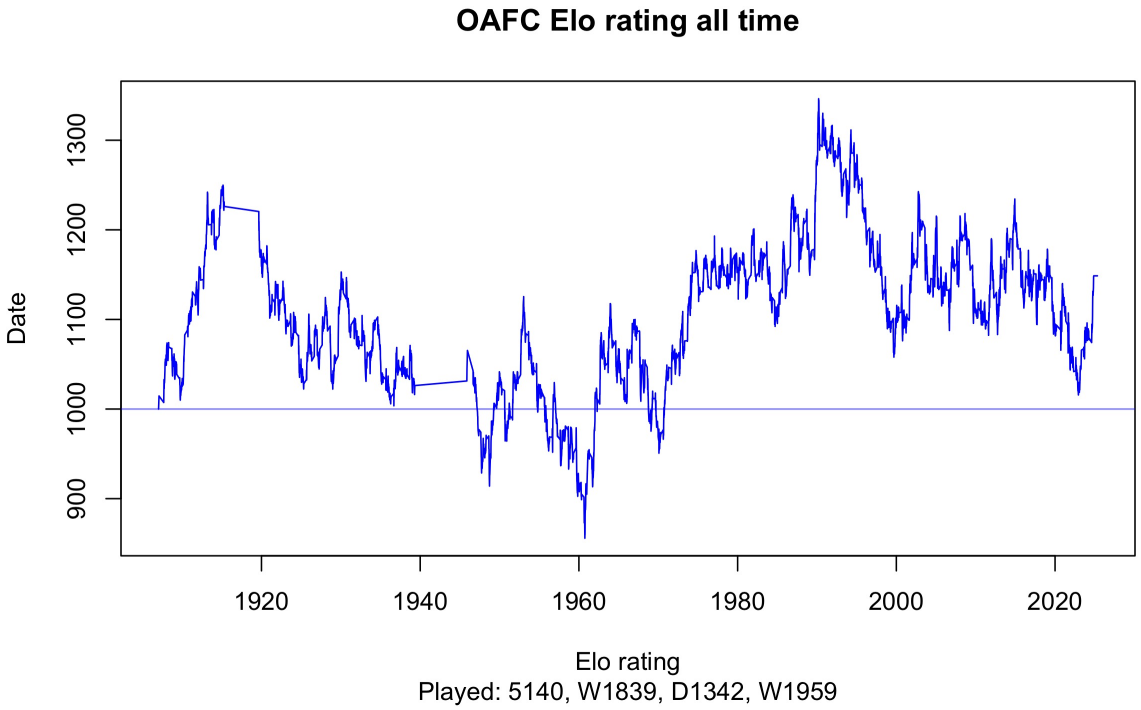


Figure 6: Oldham Athletic's Elo rating, January 1907 through to November 2024.

FL into 'non-league' competition. At that point, the club's lowest Elo rating since the early 1970s was recorded, almost resetting back to the 1000 starter value that all clubs are given.

Thus a history of over 5,000 football matches in league and cup is charted, without recourse to the league tables that nonetheless heavily dictated fortunes over that time. Elo ratings allow a comparison of club fortunes through time. There is an argument that the increase over time in the number of clubs in England and Scottish football has inflated Elo ratings, and hence Oldham's successes of the 1980s and 1990s get a higher Elo rating than in the 1910s. Conversely, though, the club played many more matches to achieve success in the 1980s and 1990s, usually playing 50 or more games compared to closer to 40 in the earlier period. It is also plausible that teams were of a higher quality at the top of the English game in the 1990s relative to the 1910s, reflecting the accumulated learning and development within the sport over this time. This could plausibly account for the difference between the two periods in terms of Elo ratings rather than necessarily being rating inflation.

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